

# Arnab Mandal

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## EDUCATION

### Shiv Nadar University

B.Tech in Computer Science Engineering (8.46/10.0)

Noida, Delhi-NCR, India

Aug 2023 - Jul 2027

### Heera Lal Public School, Class XII

Percentage : 91%

Madan Pur Dabas, Delhi-NCR, India

May 2023

### Navy Children School, Class X

Percentage : 97%

Visakhapatnam, Andhra Pradesh, India

May 2021

## INTERNSHIPS & EXPERIENCE

### AI Engineer Intern, DeepLure Research Pvt Ltd, Amravati

Dec 2025 - Jan 2026

- Optimized Llava 1.5 Pipeline serving **50+** clients by migrating inference to TensorRT alongside Triton inference server.
- Business Impact:** Achieved GPU utilization increase by **30%** and serving **8x** clients with no additional resources.

### AI Intern, Critical AI Pvt Ltd, Bengaluru

Jul 2025 - Sep 2025

- Deployed multimodal RAG pipelines (FAISS, llama-cpp, OSM, TTS, STT, MongoDB) behind FastAPI microservices.
- Business Impact:** Added **3+** features and transitioned to on-prem LLM stack, reducing inference cost by **30%**.

### AI Intern, WESEE, Ministry of Defence, New Delhi

May 2025 - Jul 2025

- Built dockerized MCP agents for SQL-based analytics and YOLOv8 pipelines (**94%** mAP@50, **68%** mAP50-95).
- Business Impact:** Cut SME report prep time by **~90%** leveraging **prompt engineering**.

## PROJECTS

### Krishi AI Sahayak - empowering farmers with smarter, data-driven agriculture

- Architected pipeline with linear regression, market-aware recommendations, and stateful, contextual AI assistant.
- Built, dockerized and deployed backend, **winning Smart SNU Hackathon** among **46** competing teams.
- Technology Stack: numpy, langchain, matplotlib, pandas

### Graph-RAG Text-to-SQL Agent for Enterprise Databases

- Engineered Text-to-SQL agent using LangGraph, tested over **43** tables and **62** relationships in a **1GB** warehouse.
- Built graph-supervised, RAG-indexed pipeline to improve multi-join query generation by **33%** on custom evals.
- Technology Stack: LangChain, LangGraph, Gemini, NetworkX, React.js

### Multilingual Multimodal Sentiment Analysis Engine

- Developed FastAPI service that applies multi-modal analysis of input, with **5** supported languages scoring **73%** accuracy.
- Architected STT service, trained transformers on text and CNNs on audio, and fused predictions via MLP.
- Technology Stack: PyTorch, scikit-learn, FastAPI, transformers

### End-to-End LLM Pretraining and Adaptation pipeline

- Implemented GPT-2 (MHA, GELU) from scratch; KV-cache raised throughput **5x** on RTX 4060.
- Fine-tuned LLaMA 3 8B using LoRA (PEFT) and quantized to Q5K-M GGUF, reducing memory footprint by **60-65%**.
- Technology Stack: PyTorch, TikToken, Unsloth, Transformers, PEFT, HuggingFace

## TECHNICAL SKILLS

|                |                                                                                          |
|----------------|------------------------------------------------------------------------------------------|
| Languages      | Python, SQL, Rust, TypeScript                                                            |
| AI & ML        | PyTorch, Transformers, LangChain, LangGraph, LlamaIndex, CrewAI, Optuna, vLLM            |
| DevOps & MLOps | FastAPI, Kubernetes, Jenkins, Docker, MLflow, DVC, Comet-ML, Triton, TensorRT, LangSmith |
| Cloud          | GCP, AWS, Azure                                                                          |

## ACHIEVEMENTS

- Smart SNU Hackathon'25 Winner:** Led a 6-member team to build and deploy the Krishi AI Sahayak app.
- LeetCode Knight:** Achieved a **1,894** peak contest rating and solved **300+** DSA problems.
- Dean's list: Monsoon'23**
- Semifinalist speaker at SMVIT Parliamentary Debate'24**